

VIHAAN RAJAGOPAL

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EDUCATION

Edison High School

Unweighted GPA: 4.20 | Weighted GPA: 5.93 | SAT: 1540 (Superscore)

Relevant Coursework: AP Biology, AP Chemistry, AP US History, AP Psychology

2028 Expected

Edison, NJ

PROFESSIONAL EXPERIENCE

Rutgers WINLAB Summer Internship advised by Prof. Jorge Ortiz —TeLLMe: Grounded Query Interface, NSF Center for Smart

Streetscapes (CS3), *Research Intern*

July 2026 – Present

- Selected as **one of 8 high school** interns across four research groups building an agentic stack that turns a **natural-language prompt into a verified, deployable distributed application**; continuing on the project through the academic year at the **invitation of the faculty lead**
- Built the **frontend for TeLLMe**, the policy layer that determines which operations and data an agent may use before any planning begins (Next.js, React, TypeScript, Tailwind)
- Built the API integration layer** binding live smart room sensors and the CARLA city simulator into a single normalized, timestamped context format, letting **one query pipeline run over both physical and simulated environments**
- Implemented **server-side proxy routes** that validate and privacy-filter every browser-facing response, keeping raw video, identities, and backend addresses **off the client**
- Expanded the verified-template library behind the planner's deterministic fast path, authoring and stress-testing templates for previously unseen query types; contributed to a group effort that **cut runtime from ~20 minutes to 20–30 seconds with no loss of model-checking coverage**.

VEX Robotics Team Member, *Engineering, Programming & Systems Optimization*

August 2025 – Present

- Placed **16th** out of 100+ teams at Regional competition
- Improved autonomous routine efficiency by **~30%** (task completion time) through algorithm tuning and iterative testing
- Optimized drivetrain control and mechanical performance, increasing movement precision and consistency during matches
- Collaborated across a 5–10 member engineering team, integrating software, hardware, and control systems

Lockheed Martin CyberQuest Participant, *Cybersecurity, Reverse Engineering & Systems Analysis*

March 2026

- Ranked **5th in the state** (top ~25%) among competitive teams
- Solved **10+** cybersecurity challenges involving reverse engineering, Linux systems, and network analysis
- Applied binary analysis, debugging, and system exploitation techniques under time-constrained conditions
- Strengthened understanding of OS internals, memory structures, and security vulnerabilities

Blue Ocean Entrepreneurship Competition, Software Developer, *Product Engineering & Backend Development*

2026

- Selected **Top 1,000 worldwide** (out of 5,000+ submissions)
- Built backend logic for a smart faucet attachment aimed at reducing micro-leaks and improving water efficiency
- Designed data flow and system logic to support real-time monitoring and leak detection
- Contributed to a sustainability-focused solution with potential to reduce household water waste **by 10–20%**

LEADERSHIP & PROJECTS

RipRead — Rip Current Risk Monitoring App, *Independent Research & Full-Stack Development*

Sept 2026 – Present

- Building a web app that pairs computer-vision analysis of beach camera footage with archived NWS surf-zone forecasts, targeting rip current risk **during unguarded hours** on NJ beaches
- Consulted **10+ domain experts** — coastal scientists at the U.S. Army Corps of Engineers, UC Santa Cruz, Monmouth University and NY Sea Grant, plus Long Branch Ocean Rescue — securing **2,700+ frames** of camera footage and an expert-labeled test set
- Trained and evaluated a segmentation model (YOLO11s-seg); built an evaluation harness that **matched the dataset authors' reference scripts exactly** and ran **pre-registered experiments**, including a rescue-day vs. control-day comparison
- Identified a **generalization gap** on expert-labeled data: the model found ~2 in 3 rips on familiar cameras but ~1 in 9 on unfamiliar ones, now guiding the next stage of development
- Automated NOAA forecast and buoy archiving with GitHub Actions; deployed a live web app with automated post-deploy checks (Python, JavaScript, Leaflet, ONNX Runtime Web, Cloudflare Pages)

- Authored **10+** technical articles on engineering, computing, and emerging technologies
- Translated complex topics (e.g., AI, cybersecurity) into accessible explanations for a student readership of **1,000+**
- Strengthened ability to communicate research ideas clearly—supporting future academic and lab work
- Assisted **50+** new student writers to start writing and get into the workflow.

Cybersecurity & Cloud Computing (Independent Study), *Hands-On Labs & Certification Track*

2024 – Present

- Completed **25+ hands-on labs** across Hack The Box and TryHackMe
- **Actively pursuing** AWS certifications; studying cloud architecture, distributed systems, and security principles
- Developed practical skills in penetration testing, system hardening, and attack simulation
- Built and tested real-world scenarios involving Linux systems, networking, and vulnerability analysis

Violinist and Student Teacher, *Performance, Mentorship, and Student Instruction*

2018 – Present

- Trained in Indian classical violin for **8+ years**, developing advanced technique, improvisation (raga), and music theory knowledge
- Accompanied vocalists at paid performances **across the tri-state area**, demonstrating real-time musical adaptation and collaboration
- Worked with diverse artists in live settings, strengthening pattern recognition, timing, and auditory analysis skills
- Taught **5–10 students** through structured one-on-one lessons, improving technical proficiency and foundational music theory
- Designed individualized lesson plans and practice strategies, fostering measurable student progress and performance readiness

SKILLS & INTERESTS

- **Technical Skills:** Python, Java, Linux, HTML, experience with LLM's and AI.
- **Technical Interests & Learning:** Cybersecurity (hands-on labs, system security, penetration testing concepts) Cloud Computing (foundations in AWS and distributed systems), Problem Solving & Systems Thinking (applying computational methods to real-world problems), developing LLMs within programs to increase convenience, implementing Smart Room Spaces in a typical living environment.
- **Certifications & Ongoing Learning:** Actively pursuing AWS certifications, hands-on cybersecurity training through Hack The Box and TryHackMe, exploring possibilities of AI and LLM's and implementing technical instances of them.
- **Hobbies:** Indian Classical Violin, Gaming, Basketball, Technology & Innovation, Communication & Teaching